

ADVANCED JAVA COURSE CONTENT

Duration- 6-8 Weeks

WEEK - 01

Introduction to Advanced Java & JDBC

- Overview of Java SE vs Java EE
- Introduction to JDBC (Java Database Connectivity)
- JDBC Drivers and architecture
- CRUD operations using JDBC
- Batch processing, Callable Statement, Transaction Management
- Practice: Connect a Java program to MySQL and perform DB operations

WEEK - 02

Java Servlets

- Introduction to Web Technologies & HTTP Protocol
- Servlet Lifecycle (init, service, destroy)
- Request & Response handling (GET, POST)
- Session Management: Cookies, URL Rewriting, HttpSession
- ServletConfig and ServletContext
- Practice: Build a login/logout web module

WEEK - 03

JavaServer Pages (JSP)

- Introduction to JSP and its lifecycle
- Scripting elements: Declaration, Expression, Scriptlet
- Directives: include, taglib, page
- JSP Implicit objects
- JSTL (Java Standard Tag Library) and EL (Expression Language)
- Practice: Create a dynamic student result page using JSP & JSTL

WEEK - 04

JavaBeans and MVC Architecture

- Introduction to JavaBeans
- Properties, Constructors, Getters/Setters
- Integrating JavaBeans with JSP
- VC Design Pattern in web applications
- Practice: Mini-project using Servlet-JSP-JavaBean MVC structure

WEEK - 05

Filters, Listeners & Annotations

- Filters in Servlets (logging, authentication, compression)
- Servlet Listeners: Context, Session, Request events
- Servlet 3.0+ Annotations (@WebServlet, @WebFilter, @WebListener)
- Practice: Implement a logging filter and session listener

WEEK - 06

JDBC Connection Pooling & Project Integration

- Connection pooling using Apache DBCP or Hikari CP
- Using Data Source for better scalability
- Deployment on Tomcat/Glass Fish server
- Practice: Convert all previous modules into a single deployable web app

OPTIONAL ADVANCED WEEKS

WEEK - 07

Introduction to Spring Framework (Bonus)

- Overview of Spring Framework
- Dependency Injection and Inversion of Control
- Spring MVC Basics
- Practice: Convert Servlet-JSP project to Spring MVC

Capstone Project & Assessment

- Capstone Project: E.g., Online Library, Attendance Tracker, Mini eCommerce
- Project Presentation
- Code Review and Assessment
- Feedback & Certificate

Technologies & Tools Used

- Languages: Java, SQL, HTML/CSS (basic)
- Databases: MySQL, PostgreSQL

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PASSION FOR INTEGRATED SOLUTIONS